

Keyush Shah

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EDUCATION

University of Pennsylvania (School of Engineering and Applied Science)

Philadelphia, PA

Master of Science in Engineering (MSE) in Data Science; GPA 3.9/4.0

Aug 2023 – May 2025

Coursework: Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception

EXPERIENCE

Software Engineer, [Omnicom](#) (NYC, New York)

Aug 2025 - Present

▪ Synthetic Audiences:

1. Architected a RAG-augmented product using **Next.js + FastAPI on Vercel/Railway** enabling pharma clients to simulate patient focus groups - interviewing AI-grounded personas & **reducing 2-4 weeks of research to minutes**.
2. Researched and prototyped a statistical **Gaussian traversal** inference-time persona generator as a clustering-free alternative for data-scarce therapeutic domains – unlocking ad-hoc research via a **sub-second inference call**.
3. Designed an HCP lens & ran a controlled A/B study -> **+10 pts on all 5 dimensions** vs patient lens (10x noise floor).

▪ Brand Intelligence Compiler (Knowledge Base):

1. Engineered a Claude based multi-step agent that orchestrated **routing -> retrieval -> tool-use answering** over an internal pharma knowledge base that **eliminated fabricated claims** across 4 client projects.
2. Designed an offline **eval harness** monitoring a north star GRR + diagnostic LLM judge flagging quality regressions.

▪ Led end-to-end migration of a multi-component ML Pipeline from AzureML to Databricks by re-architecting LLM Integration, data storage and maintaining orchestration in Spark Job DAGs unifying pipeline governance.

▪ Reduced CI/CD pipelines runtime by **12x** by migrating from Pipenv to **UV package manager**, configuring private package indexes and benchmarking Docker containers - packages installation **slashed to 12s across backend services**.

AI Researcher, [Penn Medicine - Computational Social Listening Lab](#) (PA, USA)

Dec 2024 – Aug 2026

IH Risk Model ((Part Time Research) [[CODE LINK](#)])

- Pioneered an ML hernia model using XGBoost, nested-CV and MLflow-tracked 21 ablations reducing model selection time and implementing SHAP based dilution corrections for feature importances and **raising AUC by 30% to 0.81**.
- Engineered a few shot LLM extraction pipeline on Azure OpenAI using a worker concurrency runner over ~1mm op notes (1.2M API calls, 9B tokens), hitting 96-97% field accuracy while **prompt caching cut cost by 48% (\$31K to \$16K)**.
- Built a fault tolerant multi-agent orchestration system using Claude Agent SDK featuring **LLM-as-judge** observability, automated retry logic and **FastAPI + WebSocket web interface** slashing pipeline runtime from **45 to 10 min**.
- Integrated [vLLM](#) & [LangChain](#) for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.

Data Science Intern (Full-time), [Universal Media](#) (PA, USA)

May 2024 – Aug 2024

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure.
- Drove product insights by building **Mixed Media time series models**, analyzing channel impacts on media diversity metrics.

Assistant Manager (Full-time), [IIFL Finance Ltd](#)

Apr 2022 – July 2023

- **Digital Adoption:** Led a product initiative to identify digitally savvy customers – engineered features & built ADF pipelines to track campaign behavior. Trained an ML model in AzureML and exposed as a REST endpoint driving disbursement by ~50%.

SELECTED PROJECTS

▪ Particle (2025):

- Built a context-aware chatbot for product and order support using FastAPI, LangChain, and Pinecone, implementing **RAG for product queries and tool-based logic with LLM fallback** for transactions. [\[Link\]](#)
- Orchestrated a multi-turn memory enabled >95% accurate follow-up handling and improved support efficiency by 40%.

- **Ride Duration Prediction (2025):** Developed a production-ready ML pipeline to predict NYC taxi ride durations, using **Airflow** for orchestration and **MLflow** for experiment tracking. Achieved **~30% RMSE reduction** via automated hyperparameter tuning & designed modular, reproducible workflows to simulate real-world deployment as a web service via **Flask**. [\[Link\]](#)

TECHNICAL SKILLS

Programming Languages: Python, Go, C/C++, SQL, JavaScript, Typescript, R programming

ML Libraries/Frameworks: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing

Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI

Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), GCP (BigQuery), Kafka, Airflow, DBT, PySpark

Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD